

HR Analytics

Employee Attrition & Performance Analysis

Executive HR Analytics Dashboard

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Executive Summary

Project Overview

The IBM HR Analytics Dashboard was developed using Microsoft Excel to transform raw HR data into actionable business insights.

This interactive dashboard enables HR teams and business leaders to monitor workforce performance, analyze employee attrition, evaluate salary trends, and support data-driven decision-making through dynamic KPIs, Pivot Tables, Pivot Charts, Power Query, and Slicers.

Project Snapshot

Project Type: HR Analytics Dashboard

Industry: Human Resources

Tool: Microsoft Excel

Dataset: IBM HR Analytics

Records: 1,470 Employees

Dashboard Type: Interactive Executive Dashboard

Key Features

Interactive KPIs

Dynamic Slicers

Pivot Charts

Power Query

Executive Dashboard

Business Insights

Business Problem

Organizations often struggle to understand why employees leave, which departments experience higher attrition, and how factors such as salary, overtime, job satisfaction, and experience influence workforce stability.

Traditional HR reports are static and time-consuming, making it difficult for decision-makers to identify trends quickly and take proactive actions.

An interactive HR Analytics Dashboard helps transform employee data into meaningful insights, enabling faster, data-driven workforce decisions.

Business Challenges

- High Employee Attrition
- Manual Reporting
- Poor Workforce Visibility
- Time-Consuming Analysis
- Limited Decision Support
- Lack of Interactive Insights

Business Impact

- **Faster HR Reporting**
- **Better Employee Retention**
- **Improved Workforce Planning**
- **Smarter Business Decisions**

Business Objectives

The primary objective of this project is to develop an interactive HR Analytics Dashboard that enables organizations to monitor workforce performance, understand employee attrition patterns, and support strategic HR decision-making through data-driven insights.

Key Objectives

- Monitor Workforce
- Analyze Employee Attrition
- Evaluate Salary Trends
- Measure Job Satisfaction
- Track Employee Experience
- Support HR Decision Making

Expected Business Outcomes

- Improve employee retention, ● Enable faster HR reporting, ● Identify workforce trends, ● Support strategic planning
- Enhance business decisions

Dataset Overview

Dataset Information

Dataset Name: IBM HR Analytics Employee Attrition & Performance

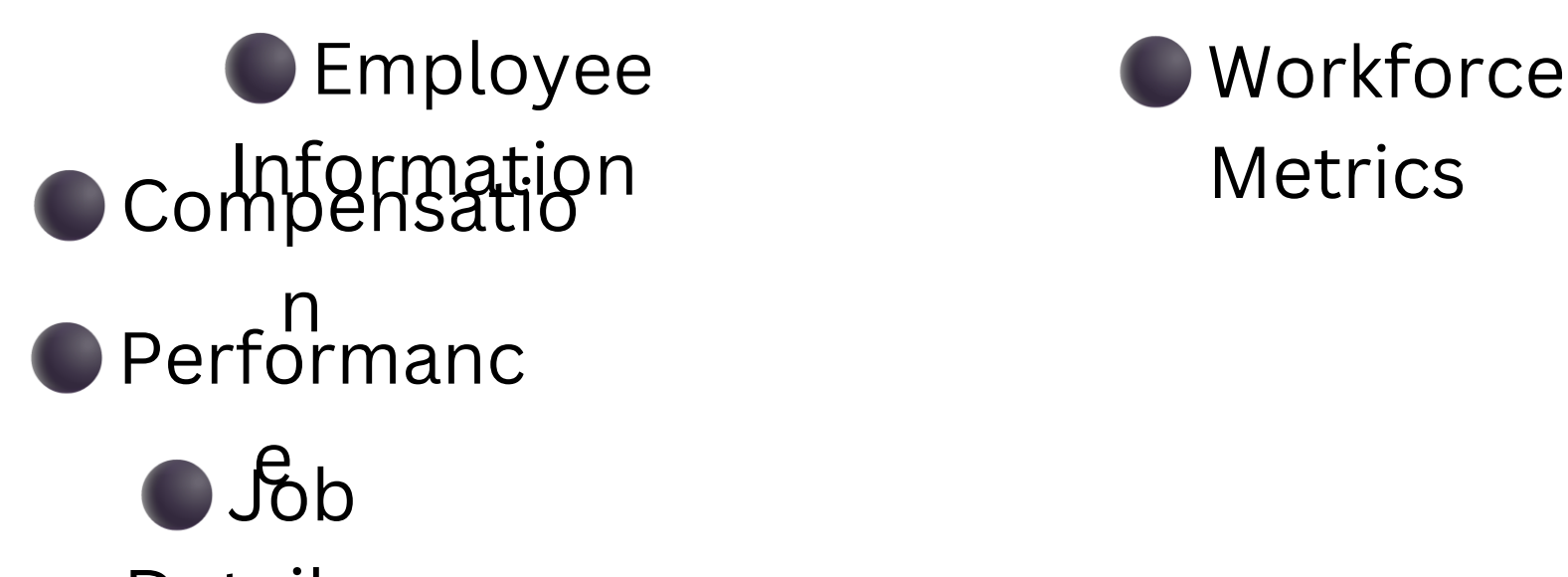
Source: IBM HR Analytics Dataset

Industry: Human Resources

Total Records: 1,470 Employees

Total Columns: 35 Features

Data Categories



Major Columns Used

- Employee
- Ag Number
- Gender
- Departmen
- Job
- Role
- Monthly
- Income
- Business Travel
- Overtime
- Job
- Satisfaction
- Years at
- Company
- Attrition

Project Scope

This project focuses on analyzing workforce data to understand employee attrition, workforce demographics, compensation, experience, and job satisfaction.

The dashboard provides an executive-level overview using interactive visualizations and dynamic filtering to support HR managers and business leaders.

Scope Includes

- Data Cleaning
- Power Query
- Pivot Tables
- Pivot Charts
- Interactive Dashboard
- Business Insights

Project Deliverables

- Interactive HR Dashboard
- Executive KPI Cards
- Department Analysis
- Job Role Analysis
- Attrition Analysis
- Business Recommendations

Data Cleaning & Preparation

The raw HR dataset was reviewed and prepared in Microsoft Excel to ensure data consistency, accuracy, and usability before analysis.

Data Preparation Steps

01 — Data Inspection

Reviewed columns, data types, and overall dataset structure.

02 — Data Quality Check

Checked for missing values, duplicates, and inconsistent entries.

03 — Data Transformation

Prepared fields and standardized data for analysis.

04 — Feature Validation

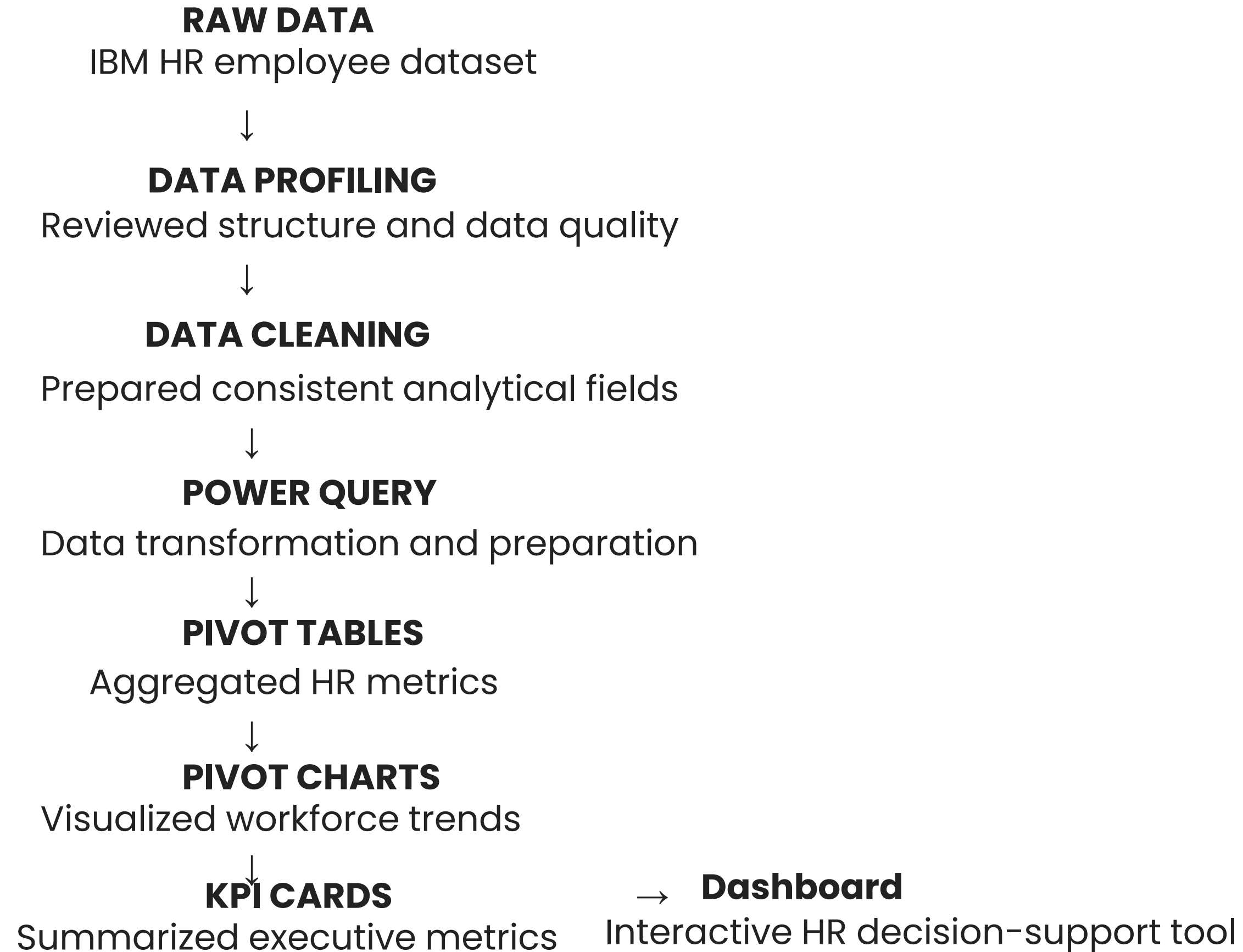
Validated key HR metrics such as Attrition, Income, Age, and Years at Company.

05 — Analysis Ready

Prepared the final dataset for Pivot Tables and dashboard reporting.

Raw HR Data → Clean Data → Analysis-Ready Dataset

Analytics Workflow



Dashboard Development

01 — KPI Layer

- Total Employees
- Attrition Count
- Attrition Rate
- Average Age
- Average Monthly Income
- Average Years at Company

02 — Analysis Layer

- Department
- Gender
- Job Role
- Salary
- Experience
- Job Satisfaction
- Overtime

03 — Interaction Layer

- Department Slicer
- Gender Slicer
- Job Role Slicer
- Education Field
- Business Travel
- Attrition

Clean • Interactive • Insight-Driven

The dashboard was designed to provide management with a quick, interactive view of workforce performance and employee attrition

Executive HR Analytics Dashboard



An interactive Excel dashboard designed to monitor workforce, attrition, compensation, experience, and employee satisfaction.

Key Business Insights

What the data reveals about workforce and employee attrition

Insight 1 — Workforce Distribution

Finding- Research & Development represents the largest employee population, followed by Sales and Human Resources.

Business Impact- Retention and workforce planning should prioritize larger employee groups while maintaining adequate staffing across critical functions.

Insight 2 — Employee Attrition

Finding- Employee attrition is concentrated within specific departments and job roles rather than being evenly distributed across the workforce.

Business Impact- HR can focus targeted retention strategies on high-attrition departments and roles instead of applying one approach to all employees.

Insight 3 — Overtime

Finding- Employees working overtime show a stronger association with employee attrition than employees who do not work overtime.

Business Impact- Monitoring workload and overtime can help reduce employee burnout and improve workforce retention.

Insight 4 — Early-Career Employees

Finding- Employees with relatively short tenure represent an important segment for attrition monitoring.

Business Impact- Stronger onboarding, mentoring, and career-development programs can improve early-stage employee retention.

Insight 5 — Job Satisfaction

Finding- Employee satisfaction varies across workforce segments, creating opportunities for targeted engagement initiatives.

Business Impact- Regular employee feedback and satisfaction monitoring can help identify potential retention risks earlier.

Insight 3 — Overtime

Finding- Employees working overtime show a stronger association with employee attrition than employees who do not work overtime.

Business Impact- Monitoring workload and overtime can help reduce employee burnout and improve workforce retention.

Priority Areas for HR

Priority 01

Employee Attrition Identify high-risk departments and job roles.

Priority 02

Overtime & Workload Monitor excessive overtime and workload pressure.

Priority 03

Employee Experience Track job satisfaction and engagement indicators.

Priority 04

Career Development Strengthen growth opportunities for employees with shorter tenure.

Data-driven workforce monitoring can help HR move from reactive reporting to proactive employee-retention strategies.

Management Recommendations

Turning workforce insights into actionable HR strategies

Recommendation 01 — Strengthen Retention

Insight Attrition is concentrated within specific employee groups.

Recommendation Develop targeted retention programs for high-attrition departments and job roles.

Expected Impact Lower avoidable employee turnover

Recommendation 02 — Manage Overtime

Insight Overtime is associated with higher employee attrition.

Recommendation Monitor overtime patterns and rebalance workloads where sustained overtime is observed.

Expected Impact Reduced burnout and improved retention

Recommendation 03 — Improve Career Growth

Insight Employees with shorter tenure require closer retention attention.

Recommendation Strengthen onboarding, mentoring, training, and career-development opportunities.

Expected Impact Higher employee engagement and longer tenure

 Recommendation 04 — Monitor Employee Experience

Insight Job satisfaction varies across workforce segments.

Recommendation Introduce regular employee feedback and satisfaction monitoring programs.

Expected Impact Earlier identification of retention risks

HR Action Framework

Turning workforce insights into actionable HR strategies

01 — Identify

Use workforce data to identify high-risk employee segments.



02 — Intervene Apply targeted retention, engagement, and workload strategies.



03 — Monitor Track attrition and workforce KPIs to measure improvement over time.

Data → **Action** → **Measurement**

A continuous analytics-driven approach can help HR teams move from reactive reporting to proactive workforce management.

Expected Business Impact

Workforce Planning

Better visibility into workforce composition

Attrition Management

More targeted employee-retention strategies

Employee Experience

Improved understanding of workforce engagement

Decision Support

Faster and more data-driven HR decisions

Technical Skills Demonstrated

Technical Skills Demonstrated

01 — Microsoft Excel

Advanced Excel

Data Analysis

↓
Dashboard Development

02 — Power Query

Data Cleaning

Transformation

Preparation

03 — Pivot Tables

Data Aggregation

KPI Calculation

Workforce Analysis

04 — Pivot Charts

Data Visualization

Trend Analysis

Comparison

05 — Slicers

Interactive Filtering
Dynamic Analysis

06 — KPI Design

Executive Metrics
Performance Monitoring

07 — Dashboard Design

Executive Layout
Visual Storytelling

08 — Business Analysis

Insights
Recommendations
Decision Support

End-to-End Analytics Workflow

From raw employee data to actionable HR insights

01 — Data

IBM HR Analytics Dataset



02 — Clean

Data quality checks



03 — Transform

Power Query



04 — Analyze

Pivot Tables



05 — Visualize

Pivot Charts + KPIs



06 — Insight

Attrition & Workforce Analysis



07 — Action

HR Recommendations

Thank You

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